

Problem Set 10:

Submission: 12:00, Friday, June 19, 2026; Solutions: Friday, July 3, 2026

Problem 1 Retarded Green Functions in 3+1 and 2+1 Dimensions and Huygens' Principle

In the lecture, the following expression for the retarded Green function of the electromagnetic field was derived (Eq. (62) in the script):

$$\tilde{G}_{\text{ret}}(t, \mathbf{k}) = \frac{1}{2\pi} \frac{1}{\sqrt{2\pi}} \frac{\sin(kt)}{k}. \quad (1)$$

(1.a) Derive the Green function in *three*-dimensional position space,

$$G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') = \begin{cases} 0, & t < t', \\ \frac{\delta(t - t' - |\mathbf{x} - \mathbf{x}'|/c)}{4\pi |\mathbf{x} - \mathbf{x}'|}, & t > t', \end{cases} \quad (2)$$

by explicitly carrying out the Fourier transformation.

We now consider the two-dimensional case. A direct integration as in part (a) is still a viable approach, but the integrals that occur are more difficult to solve. Instead, we want to use our result from three-dimensional position space in a clever way in order to derive the solution in two dimensions.

Specifically, we recall that

$$(c^{-2}\partial_t^2 - \nabla^2) G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') = \delta(t - t')\delta^{(3)}(\mathbf{x} - \mathbf{x}'), \quad (3)$$

defines the Green function.

We now introduce

$$G_{\text{ret}}^{(2)}(t, \mathbf{x}_{\perp}; t', \mathbf{x}'_{\perp}) = \int_{-\infty}^{\infty} dz G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}'), \quad (4)$$

with $\mathbf{x}_{\perp} = (x, y)^T$. One quickly verifies that the result does not depend on the z -component of $\mathbf{x}'$ on the right-hand side of this equation. Hence $G_{\text{ret}}^{(2)}(t, \mathbf{x}_{\perp}; t', \mathbf{x}'_{\perp})$ is well-defined.

(1.b) Show that $G_{\text{ret}}^{(2)}$ is the Green function of the wave equation in two spatial dimensions:

$$(c^{-2}\partial_t^2 - \partial_x^2 - \partial_y^2) G_{\text{ret}}^{(2)}(t, \mathbf{x}_{\perp}; t', \mathbf{x}'_{\perp}) = \delta(t - t')\delta^{(2)}(\mathbf{x}_{\perp} - \mathbf{x}'_{\perp}). \quad (5)$$

Argue why the boundary term that occurs may be neglected.

(1.c) Now explicitly carry out the integral over z in order to derive

$$G_{\text{ret}}^{(2)}(t, \mathbf{x}_{\perp}; t', \mathbf{x}'_{\perp}) = \frac{c}{2\pi} \frac{\Theta(t - t' - |\mathbf{x}_{\perp} - \mathbf{x}'_{\perp}|/c)}{\sqrt{c^2(t - t')^2 - |\mathbf{x}_{\perp} - \mathbf{x}'_{\perp}|^2}} \quad (6)$$

Here Θ is the Heaviside function.

Hint: Recall the identity $\delta(f(z)) = \sum_i \frac{\delta(z - z_i)}{|f'(z_i)|}$, where the sum runs over the simple zeros z_i of the function f .

(1.d) Now interpret the result for $G_{\text{ret}}^{(2)}$ physically by establishing the connection to Huygens' principle in 3+1 dimensions. Which points in spacetime contribute to $G_{\text{ret}}^{(2)}$? Compare with the three-dimensional case.

Solution:

(1.a) As prescribed, we start from

$$\tilde{G}_{\text{ret}}(t, \mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \frac{\sin(kt)}{k}, \quad k = |\mathbf{k}|. \quad (7)$$

For fixed $\mathbf{x} \in \mathbb{R}^3$ and $t > 0$, we obtain G_{ret} by Fourier transformation:

$$\begin{aligned} G_{\text{ret}}(t, \mathbf{x}) &= \frac{1}{(2\pi)^{3/2}} \int d^3\mathbf{k} \tilde{G}_{\text{ret}}(t, \mathbf{k}) \exp(i\mathbf{k} \cdot \mathbf{x}) \\ &= \frac{1}{8\pi^3} \int_0^\infty dk \int_0^{2\pi} d\varphi \int_0^\pi d\theta k \sin(kt) \sin\theta \exp(ik|\mathbf{x}| \cos\theta) \\ &= \frac{1}{4\pi^2} \int_0^\infty dk k \sin(kt) \int_0^\pi d\theta \sin\theta \exp(ik|\mathbf{x}| \cos\theta) \\ &= \frac{1}{4\pi^2} \int_0^\infty dk k \sin(kt) \int_{-1}^1 du \exp(ik|\mathbf{x}|u) \\ &= \frac{1}{4\pi^2} \int_0^\infty dk k \sin(kt) \frac{1}{ik|\mathbf{x}|} (e^{ik|\mathbf{x}|} - e^{-ik|\mathbf{x}|}) \\ &= -\frac{1}{8\pi^2|\mathbf{x}|} \int_0^\infty dk (e^{ik|\mathbf{x}|} - e^{-ik|\mathbf{x}|}) (e^{ikt} - e^{-ikt}) \\ &= -\frac{1}{8\pi^2|\mathbf{x}|} \int_0^\infty dk (e^{ik(|\mathbf{x}|+t)} + e^{-ik(|\mathbf{x}|+t)} - e^{ik(|\mathbf{x}|-t)} - e^{-ik(|\mathbf{x}|-t)}) \\ &= -\frac{1}{8\pi^2|\mathbf{x}|} \int_{-\infty}^\infty dk (e^{ik(|\mathbf{x}|+t)} - e^{ik(|\mathbf{x}|-t)}) \\ &= -\frac{1}{4\pi|\mathbf{x}|} (\delta(t + |\mathbf{x}|) - \delta(t - |\mathbf{x}|)) \\ &= \frac{\delta(t - |\mathbf{x}|)}{4\pi|\mathbf{x}|}. \end{aligned} \quad (8)$$

We have used that

$$\int_{-\infty}^{\infty} dk e^{ika} = 2\pi\delta(a), \quad (9)$$

as well as the fact that we consider $t > 0$, so that $\delta(t + |\mathbf{x}|) = 0$ holds.

By translation in $(t, \mathbf{x})$, it follows immediately that

$$G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') = \begin{cases} 0, & t < t', \\ \frac{\delta(t - t' - |\mathbf{x} - \mathbf{x}'|/c)}{4\pi|\mathbf{x} - \mathbf{x}'|}, & t > t'. \end{cases} \quad (10)$$

Here we have reintroduced the factor c ; in the intermediate calculation, we set $c = 1$.

(1.b) We apply

$$c^{-2}\partial_t^2 - \partial_x^2 - \partial_y^2 \quad (11)$$

to

$$G_{\text{ret}}^{(2)}(t, \mathbf{x}_{\perp}; t', \mathbf{x}'_{\perp}) = \int_{-\infty}^{\infty} dz G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}'). \quad (12)$$

In doing so, we write

$$\mathbf{x} = (\mathbf{x}_{\perp}, z), \quad \mathbf{x}' = (\mathbf{x}'_{\perp}, z'). \quad (13)$$

Then

$$\begin{aligned} & (c^{-2}\partial_t^2 - \partial_x^2 - \partial_y^2) \int_{-\infty}^{\infty} dz G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') \\ &= \int_{-\infty}^{\infty} dz (c^{-2}\partial_t^2 - \nabla^2 + \partial_z^2) G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') \\ &= \int_{-\infty}^{\infty} dz \delta(t - t') \delta^{(2)}(\mathbf{x}_{\perp} - \mathbf{x}'_{\perp}) \delta(z - z') + \int_{-\infty}^{\infty} dz \partial_z^2 G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') \\ &= \delta(t - t') \delta^{(2)}(\mathbf{x}_{\perp} - \mathbf{x}'_{\perp}) + \int_{-\infty}^{\infty} dz \partial_z^2 G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}'). \end{aligned} \quad (14)$$

The last term is to be understood in the sense of distributions. To this end, let $\phi(t, \mathbf{x}_{\perp})$ be a test function with compact support and define

$$T_{\phi}(z) = \int_{\mathbb{R}} dt \int_{\mathbb{R}^2} d^2\mathbf{x}_{\perp} G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') \phi(t, \mathbf{x}_{\perp}). \quad (15)$$

Since $G_{\text{ret}}^{(3)}$ is nonzero only for

$$t - t' = \frac{1}{c} \sqrt{|\mathbf{x}_{\perp} - \mathbf{x}'_{\perp}|^2 + (z - z')^2}, \quad (16)$$

and since ϕ has compact support in $(t, \mathbf{x}_\perp)$, $T_\phi(z)$ also has compact support in z . Therefore,

$$\left\langle \int_{-\infty}^{\infty} dz \partial_z^2 G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}'), \phi \right\rangle = \int_{-\infty}^{\infty} dz \partial_z^2 T_\phi(z) = 0. \quad (17)$$

Thus the boundary term vanishes, and what remains is

$$(c^{-2} \partial_t^2 - \partial_x^2 - \partial_y^2) G_{\text{ret}}^{(2)}(t, \mathbf{x}_\perp; t', \mathbf{x}'_\perp) = \delta(t - t') \delta^{(2)}(\mathbf{x}_\perp - \mathbf{x}'_\perp). \quad (18)$$

(1.c) For $t < t'$,

$$G_{\text{ret}}^{(3)}(t, \mathbf{x}; t', \mathbf{x}') = 0 \quad (19)$$

and therefore also

$$G_{\text{ret}}^{(2)}(t, \mathbf{x}_\perp; t', \mathbf{x}'_\perp) = 0. \quad (20)$$

We now set

$$\tau = t - t' > 0, \quad \rho = |\mathbf{x}_\perp - \mathbf{x}'_\perp|. \quad (21)$$

After a shift $z - z' \mapsto z$, we have

$$\begin{aligned} G_{\text{ret}}^{(2)}(t, \mathbf{x}_\perp; t', \mathbf{x}'_\perp) &= \int_{-\infty}^{\infty} dz \frac{\delta\left(t - t' - \sqrt{\rho^2 + z^2}/c\right)}{4\pi\sqrt{\rho^2 + z^2}} \\ &= \int_{-\infty}^{\infty} dz \frac{\delta\left(\tau - \sqrt{\rho^2 + z^2}/c\right)}{4\pi\sqrt{\rho^2 + z^2}}. \end{aligned} \quad (22)$$

Let

$$f(z) = \tau - \frac{\sqrt{\rho^2 + z^2}}{c}. \quad (23)$$

Then

$$f(z) = 0 \iff z = z_\pm = \pm \sqrt{c^2 \tau^2 - \rho^2}. \quad (24)$$

Thus zeros exist only for $c\tau > \rho$. Moreover,

$$f'(z_\pm) = -\frac{z_\pm}{c\sqrt{\rho^2 + z_\pm^2}} = \mp \frac{\sqrt{c^2 \tau^2 - \rho^2}}{c^2 \tau}. \quad (25)$$

With

$$\delta(f(z)) = \sum_{\sigma=\pm} \frac{\delta(z - z_\sigma)}{|f'(z_\sigma)|} \quad (26)$$

it follows that

$$\begin{aligned} G_{\text{ret}}^{(2)}(t, \mathbf{x}_\perp; t', \mathbf{x}'_\perp) &= \frac{1}{4\pi} \sum_{\sigma=\pm} \frac{1}{|f'(z_\sigma)|} \frac{1}{\sqrt{\rho^2 + z_\sigma^2}} \Theta(c\tau - \rho) \\ &= \frac{1}{4\pi} \frac{c^2\tau}{\sqrt{c^2\tau^2 - \rho^2}} \left(\frac{1}{c\tau} + \frac{1}{c\tau} \right) \Theta(c\tau - \rho) \\ &= \frac{c}{2\pi} \frac{\Theta(c\tau - \rho)}{\sqrt{c^2\tau^2 - \rho^2}}. \end{aligned} \quad (27)$$

Hence we obtain

$$G_{\text{ret}}^{(2)}(t, \mathbf{x}_\perp; t', \mathbf{x}'_\perp) = \frac{c}{2\pi} \frac{\Theta(t - t' - |\mathbf{x}_\perp - \mathbf{x}'_\perp|/c)}{\sqrt{c^2(t - t')^2 - |\mathbf{x}_\perp - \mathbf{x}'_\perp|^2}}. \quad (28)$$

Problem 2 Semicircular Waveguide

Consider a perfectly conducting waveguide with a cross section in the shape of a semicircle with radius R .

- (2.a) Determine the longitudinal fields E_z and B_z , respectively, for the TM and TE modes. Also determine the cutoff frequency of these modes.
- (2.b) Give explicit formulas for the transverse fields $\mathbf{E}_\perp$ and $\mathbf{B}_\perp$ for the lowest cutoff frequency found in part (a).

Solution:

We assume that the waveguide is infinitely long in the z -direction. Its cross-section lies in the xy -plane and is chosen to be the upper semicircular disk:

$$A = \{(x, y) : x^2 + y^2 \leq R^2, y \geq 0\}. \quad (29)$$

Equivalently, in polar coordinates in the transverse plane,

$$x = \rho \cos \varphi, \quad y = \rho \sin \varphi, \quad (30)$$

and the cross-section is given by

$$0 \leq \rho \leq R, \quad 0 \leq \varphi \leq \pi. \quad (31)$$

The conducting boundary of the cross-section consists of two parts. The flat bottom is

$$y = 0, \quad -R \leq x \leq R. \quad (32)$$

In polar coordinates, this flat bottom is described by the two radial boundary pieces

$$\Gamma_0 : \varphi = 0, \quad \Gamma_\pi : \varphi = \pi. \quad (33)$$

The curved semicircular wall is

$$x^2 + y^2 = R^2, \quad y \geq 0, \quad (34)$$

or, in polar coordinates,

$$\Gamma_R : \rho = R. \quad (35)$$

The complete conducting surface of the waveguide is obtained by translating this boundary along the z -axis. Thus every wall is parallel to the z -direction.

We use the complex convention

$$\mathbf{E}, \mathbf{B} \propto e^{ik_z z - i\omega t}. \quad (36)$$

The physical fields are obtained by taking the real part. For a perfect conductor, the tangential electric field vanishes at the wall:

$$\hat{\mathbf{n}} \times \mathbf{E} = 0. \quad (37)$$

This is the boundary condition from the lecture script, equation (181). For a waveguide, this gives the corresponding transverse boundary condition

$$\hat{\mathbf{n}} \times \mathbf{E}^T = 0 \quad \text{on } \partial A, \quad (38)$$

as in equation (205) of the script.

The transverse Laplace operator is

$$\nabla_\perp^2 = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2}{\partial \varphi^2}. \quad (39)$$

For both TE and TM modes, the relevant longitudinal field satisfies a scalar Helmholtz equation on the cross-section:

$$(\nabla_\perp^2 + k_c^2) \psi = 0, \quad (40)$$

where

$$k_c^2 = \frac{\omega^2}{c^2} - k_z^2. \quad (41)$$

Equivalently,

$$k_z^2 = \frac{\omega^2}{c^2} - k_c^2. \quad (42)$$

A propagating mode therefore requires $k_z^2 > 0$. The cut-off angular frequency is the frequency at which $k_z = 0$, hence

$$\omega_c = ck_c. \quad (43)$$

We now solve

$$(\nabla_\perp^2 + k_c^2) \psi = 0 \quad (44)$$

on the semicircular disk. We use the separation ansatz

$$\psi(\rho, \varphi) = P(\rho)\Phi(\varphi). \quad (45)$$

Substitution into the transverse Helmholtz equation gives

$$\begin{aligned} 0 &= (\nabla_{\perp}^2 + k_c^2) (P(\rho)\Phi(\varphi)) \\ &= \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) \Phi + \frac{1}{\rho^2} P \frac{d^2\Phi}{d\varphi^2} + k_c^2 P \Phi. \end{aligned} \quad (46)$$

Dividing by $P\Phi$ and multiplying by ρ^2 yields

$$\frac{\rho}{P} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + k_c^2 \rho^2 + \frac{1}{\Phi} \frac{d^2\Phi}{d\varphi^2} = 0. \quad (47)$$

The first two terms depend only on ρ , while the last term depends only on φ . Therefore both parts must be constant. We write the angular separation constant as $-m^2$, so that

$$\frac{1}{\Phi} \frac{d^2\Phi}{d\varphi^2} = -m^2. \quad (48)$$

Thus the angular equation is

$$\frac{d^2\Phi}{d\varphi^2} + m^2\Phi = 0. \quad (49)$$

The radial equation is then

$$\frac{\rho}{P} \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + k_c^2 \rho^2 - m^2 = 0. \quad (50)$$

Multiplication by P gives

$$\rho \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + (k_c^2 \rho^2 - m^2) P = 0. \quad (51)$$

Equivalently,

$$\rho^2 \frac{d^2P}{d\rho^2} + \rho \frac{dP}{d\rho} + (k_c^2 \rho^2 - m^2) P = 0. \quad (52)$$

Introducing

$$s = k_c \rho \quad (53)$$

we obtain

$$s^2 \frac{d^2P}{ds^2} + s \frac{dP}{ds} + (s^2 - m^2) P = 0. \quad (54)$$

This is Bessel's equation of order m . Hence

$$P(s) = AJ_m(s) + BY_m(s). \quad (55)$$

The point $\rho = 0$ lies in the closure of the physical domain; geometrically, it is the midpoint of the flat bottom. The electromagnetic fields must remain finite there. Since $Y_m(s)$ is singular at $s = 0$, we set $B = 0$. Thus the regular radial solutions are

$$P(\rho) = J_m(k_c \rho). \quad (56)$$

(2.a) For TM modes,

$$B_z = 0, \quad E_z \neq 0. \quad (57)$$

The longitudinal electric field E_z points in the z -direction. Since the conducting walls are obtained by translating the boundary of the cross-section along the z -axis, the z -direction is tangential to every conducting wall. Therefore the condition for a perfect conductor,

$$\hat{\mathbf{n}} \times \mathbf{E} = 0, \quad (58)$$

implies that the tangential component E_z must vanish on the entire boundary:

$$E_z = 0 \quad \text{on } \partial A. \quad (59)$$

Thus E_z satisfies a Dirichlet boundary condition. This is the Dirichlet eigenvalue problem for the functions α_n in equation (216) of the script.

We write

$$E_z(\rho, \varphi, z, t) = P(\rho)\Phi(\varphi)e^{ik_z z - i\omega t}. \quad (60)$$

First we impose the boundary condition on the flat bottom. In polar coordinates, the two pieces of the flat bottom are

$$\Gamma_0 : \varphi = 0, \quad \Gamma_\pi : \varphi = \pi. \quad (61)$$

Since E_z is tangential to the conducting wall and the tangential electric field must vanish on a perfect conductor, we have

$$E_z(\rho, 0, z, t) = 0, \quad E_z(\rho, \pi, z, t) = 0. \quad (62)$$

Since the factors $P(\rho)$ and $e^{ik_z z - i\omega t}$ do not vanish identically, these two boundary conditions imply

$$\Phi(0) = 0, \quad \Phi(\pi) = 0. \quad (63)$$

The conditions $\Phi(0) = 0$ and $\Phi(\pi) = 0$ therefore follow directly from the requirement $E_z = 0$ on the two straight conducting pieces Γ_0 and Γ_π of the flat bottom.

The angular equation has the general solution

$$\Phi(\varphi) = A \cos(m\varphi) + B \sin(m\varphi). \quad (64)$$

The condition $\Phi(0) = 0$, which follows from $E_z = 0$ on Γ_0 , gives

$$A = 0. \quad (65)$$

The condition $\Phi(\pi) = 0$, which follows from $E_z = 0$ on Γ_π , then gives

$$B \sin(m\pi) = 0. \quad (66)$$

For a nontrivial solution, we must have $B \neq 0$. Therefore

$$\sin(m\pi) = 0. \quad (67)$$

Thus

$$m = 1, 2, 3, \dots \quad (68)$$

The angular functions for TM modes are

$$\Phi_m(\varphi) = \sin(m\varphi), \quad m = 1, 2, 3, \dots \quad (69)$$

Next we impose the boundary condition on the curved wall:

$$\Gamma_R : \rho = R. \quad (70)$$

The curved wall is also parallel to the z -direction. Therefore E_z is tangential to this wall as well. The perfect-conductor condition again requires the tangential electric field to vanish, so

$$E_z(R, \varphi, z, t) = 0. \quad (71)$$

Since the angular factor $\sin(m\varphi)$ does not vanish identically, it follows that

$$P(R) = 0. \quad (72)$$

With $P(\rho) = J_m(k_c \rho)$ we obtain

$$J_m(k_c R) = 0. \quad (73)$$

Let j_{mn} be the n -th positive zero of J_m . Then

$$k_c R = j_{mn}. \quad (74)$$

Thus

$$k_c = \frac{j_{mn}}{R}. \quad (75)$$

The longitudinal TM fields are therefore

$$E_{z,mn}^{\text{TM}} = E_{0,mn} J_m \left(\frac{j_{mn} \rho}{R} \right) \sin(m\varphi) e^{ik_z z - i\omega t}, \quad (76)$$

and

$$B_{z,mn}^{\text{TM}} = 0. \quad (77)$$

The dispersion relation is

$$k_z^2 = \frac{\omega^2}{c^2} - \frac{j_{mn}^2}{R^2}. \quad (78)$$

Hence the cut-off angular frequencies of the TM modes are

$$\omega_{c,mn}^{\text{TM}} = \frac{c j_{mn}}{R}, \quad m = 1, 2, 3, \dots, \quad n = 1, 2, 3, \dots \quad (79)$$

The lowest TM cut-off is determined by

$$j_{11} \approx 3.831706, \quad (80)$$

so that

$$\omega_{c,11}^{\text{TM}} \approx \frac{3.831706 c}{R}. \quad (81)$$

For TE modes,

$$E_z = 0, \quad B_z \neq 0, \quad (82)$$

and the transverse electric field can be written in terms of the longitudinal magnetic field as

$$\mathbf{E}_\perp = -\frac{i\omega}{k_c^2} \hat{\mathbf{z}} \times \nabla_\perp B_z. \quad (83)$$

Let $\hat{\mathbf{n}}$ be the outward normal vector to the boundary curve in the xy -plane and $\hat{\mathbf{t}}$ the tangent vector to this boundary curve. Since the conducting wall is obtained by translating this boundary curve along the z -direction, the two tangential directions on the conducting wall are $\hat{\mathbf{z}}$ and $\hat{\mathbf{t}}$.

For TE modes, $E_z = 0$. Therefore the tangential electric field in the $\hat{\mathbf{z}}$ -direction vanishes automatically. The remaining perfect-conductor condition is

$$\mathbf{E}_\perp \cdot \hat{\mathbf{t}} = 0. \quad (84)$$

With

$$\mathbf{E}_\perp = -\frac{i\omega}{k_c^2} \hat{\mathbf{z}} \times \nabla_\perp B_z \quad (85)$$

this tangential component is proportional to the normal derivative of B_z . Therefore the conductor condition is equivalent to

$$\frac{\partial B_z}{\partial n} = 0 \quad \text{on } \partial A. \quad (86)$$

Thus B_z satisfies a Neumann boundary condition.

We write

$$B_z(\rho, \varphi, z, t) = P(\rho)\Phi(\varphi)e^{ik_z z - i\omega t}. \quad (87)$$

First we impose the Neumann boundary condition on the flat bottom. On the boundary piece

$$\Gamma_0 : \varphi = 0 \quad (88)$$

the outward normal points downward, i.e. in the negative y -direction. In polar coordinates this is

$$\hat{\mathbf{n}} = -\hat{\varphi} \quad \text{on } \varphi = 0. \quad (89)$$

Therefore

$$\frac{\partial}{\partial n} = -\frac{1}{\rho} \frac{\partial}{\partial \varphi} \quad \text{on } \varphi = 0. \quad (90)$$

On the boundary piece

$$\Gamma_\pi : \varphi = \pi \quad (91)$$

the outward normal again points downward, but in polar coordinates this is now $+\hat{\varphi}$. Therefore

$$\frac{\partial}{\partial n} = \frac{1}{\rho} \frac{\partial}{\partial \varphi} \quad \text{on } \varphi = \pi. \quad (92)$$

The signs are different on the two pieces, but the Neumann condition sets the normal derivative equal to zero in both cases. Hence

$$\frac{\partial B_z}{\partial \varphi} = 0 \quad \text{at } \varphi = 0, \pi. \quad (93)$$

Since

$$B_z = P(\rho)\Phi(\varphi)e^{ik_z z - i\omega t} \quad (94)$$

and $P(\rho)e^{ik_z z - i\omega t}$ does not vanish identically, the Neumann condition on the flat bottom implies

$$\Phi'(0) = 0, \quad \Phi'(\pi) = 0. \quad (95)$$

The conditions $\Phi'(0) = 0$ and $\Phi'(\pi) = 0$ therefore follow directly from $\partial B_z / \partial n = 0$ on the two straight conducting pieces of the flat bottom.

The angular equation again has the general solution

$$\Phi(\varphi) = A \cos(m\varphi) + B \sin(m\varphi). \quad (96)$$

The condition $\Phi'(0) = 0$, which follows from the Neumann condition on Γ_0 , gives

$$\Phi'(\varphi) = -mA \sin(m\varphi) + mB \cos(m\varphi), \quad (97)$$

and hence

$$\Phi'(0) = mB = 0. \quad (98)$$

For $m > 0$, this implies $B = 0$. The case $m = 0$ is included separately and gives a constant angular function. Thus the allowed angular functions are of cosine type. The second condition $\Phi'(\pi) = 0$, which follows from the Neumann condition on Γ_π , gives

$$-mA \sin(m\pi) = 0. \quad (99)$$

For a nontrivial solution, this requires

$$\sin(m\pi) = 0. \quad (100)$$

Therefore

$$m = 0, 1, 2, \dots \quad (101)$$

The angular functions for TE modes are

$$\Phi_m(\varphi) = \cos(m\varphi), \quad m = 0, 1, 2, \dots \quad (102)$$

Next we impose the Neumann boundary condition on the curved wall:

$$\Gamma_R : \rho = R. \quad (103)$$

On this boundary the outward normal is

$$\hat{\mathbf{n}} = \hat{\boldsymbol{\rho}}. \quad (104)$$

Therefore

$$\frac{\partial}{\partial n} = \frac{\partial}{\partial \rho} \quad \text{on } \rho = R. \quad (105)$$

The Neumann condition $\partial B_z / \partial n = 0$ gives

$$\left. \frac{\partial B_z}{\partial \rho} \right|_{\rho=R} = 0. \quad (106)$$

Since

$$B_z = P(\rho)\Phi(\varphi)e^{ik_z z - i\omega t} \quad (107)$$

and the angular factor does not vanish identically, it follows that

$$P'(R) = 0. \quad (108)$$

With

$$P(\rho) = J_m(k_c \rho) \quad (109)$$

we obtain

$$P'(\rho) = k_c J'_m(k_c \rho). \quad (110)$$

Therefore the boundary condition on the curved wall yields

$$J'_m(k_c R) = 0. \quad (111)$$

Let j'_{mn} be the n -th positive zero of J'_m . Then

$$k_c R = j'_{mn}, \quad (112)$$

and hence

$$k_c = \frac{j'_{mn}}{R}. \quad (113)$$

The longitudinal TE fields are therefore

$$B_{z,mn}^{\text{TE}} = B_{0,mn} J_m \left(\frac{j'_{mn} \rho}{R} \right) \cos(m\varphi) e^{ik_z z - i\omega t}, \quad (114)$$

and

$$E_{z,mn}^{\text{TE}} = 0. \quad (115)$$

The dispersion relation is

$$k_z^2 = \frac{\omega^2}{c^2} - \frac{(j'_{mn})^2}{R^2}. \quad (116)$$

Hence the cut-off angular frequencies of the TE modes are

$$\omega_{c,mn}^{\text{TE}} = \frac{c j'_{mn}}{R}, \quad m = 0, 1, 2, \dots, \quad n = 1, 2, 3, \dots \quad (117)$$

Formally, there is a constant Neumann eigenfunction with $k_c = 0$. However, it does not yield a nontrivial TE mode, since the transverse electric field is proportional to $\hat{\mathbf{z}} \times \nabla_{\perp} B_z$ and the gradient of a constant vanishes.

The smallest positive zero among the TE modes is

$$j'_{11} \approx 1.841184. \quad (118)$$

Therefore the mode with the lowest cut-off in the semicircular waveguide is the mode TE_{11} with cut-off angular frequency

$$\omega_c^{\text{lowest}} = \frac{1.841184 c}{R}. \quad (119)$$

(2.b) The lowest mode is the TE_{11} mode. Define

$$u := j'_{11} \approx 1.841184, \quad q := \frac{u}{R}. \quad (120)$$

Then

$$q = k_c, \quad \omega_c = cq. \quad (121)$$

The longitudinal magnetic field is

$$B_z = B_0 J_1(q\rho) \cos \varphi e^{ik_z z - i\omega t}. \quad (122)$$

The longitudinal electric field is

$$E_z = 0. \quad (123)$$

The dispersion relation is

$$k_z^2 = \frac{\omega^2}{c^2} - q^2. \quad (124)$$

For TE modes,

$$\mathbf{E}_\perp = -\frac{i\omega}{q^2} \hat{\mathbf{z}} \times \nabla_\perp B_z, \quad (125)$$

and

$$\mathbf{B}_\perp = \frac{ik_z}{q^2} \nabla_\perp B_z. \quad (126)$$

We calculate

$$\begin{aligned} \nabla_\perp B_z &= \hat{\rho} \frac{\partial B_z}{\partial \rho} + \hat{\varphi} \frac{1}{\rho} \frac{\partial B_z}{\partial \varphi} \\ &= B_0 e^{ik_z z - i\omega t} \left[q J_1'(q\rho) \cos \varphi \hat{\rho} - \frac{J_1(q\rho)}{\rho} \sin \varphi \hat{\varphi} \right]. \end{aligned} \quad (127)$$

Using

$$\hat{\mathbf{z}} \times \hat{\rho} = \hat{\varphi}, \quad \hat{\mathbf{z}} \times \hat{\varphi} = -\hat{\rho}, \quad (128)$$

we find

$$\mathbf{E}_\perp = -\frac{i\omega B_0}{q^2} e^{ik_z z - i\omega t} \left[\frac{J_1(q\rho)}{\rho} \sin \varphi \hat{\rho} + q J_1'(q\rho) \cos \varphi \hat{\varphi} \right]. \quad (129)$$

Thus

$$E_\rho = -\frac{i\omega B_0}{q^2} \frac{J_1(q\rho)}{\rho} \sin \varphi e^{ik_z z - i\omega t}, \quad (130)$$

and

$$E_\varphi = -\frac{i\omega B_0}{q} J_1'(q\rho) \cos \varphi e^{ik_z z - i\omega t}. \quad (131)$$

Similarly,

$$B_\rho = \frac{ik_z B_0}{q} J_1'(q\rho) \cos \varphi e^{ik_z z - i\omega t}, \quad (132)$$

and

$$B_\varphi = -\frac{ik_z B_0}{q^2} \frac{J_1(q\rho)}{\rho} \sin \varphi e^{ik_z z - i\omega t}. \quad (133)$$

At cut-off,

$$\omega = \omega_c = cq, \quad k_z = 0. \quad (134)$$

Therefore

$$B_\rho = B_\varphi = 0, \quad (135)$$

while

$$\mathbf{E}_\perp = -\frac{icB_0}{q} \left[\frac{J_1(q\rho)}{\rho} \sin \varphi \hat{\rho} + q J_1'(q\rho) \cos \varphi \hat{\varphi} \right] e^{-i\omega_c t}. \quad (136)$$

Problem 3 (Central Tutorial) Positivity and Hermiticity of the Laplace Operator

Let $V \subset \mathbb{R}^3$ be a bounded volume with smooth boundary ∂V and outward normal vector $\hat{\mathbf{n}}(\mathbf{r})$. We consider the usual scalar product $\langle \psi, \phi \rangle$ for complex-valued and sufficiently smooth functions. Consider the operator

$$\mathcal{L} = -\nabla^2 \quad (137)$$

on functions ψ with Dirichlet boundary condition

$$\psi|_{\partial V} = 0. \quad (138)$$

(3.a) Show that $\mathcal{L}$ is Hermitian, i.e. that

$$\langle \mathcal{L}\psi, \phi \rangle = \langle \psi, \mathcal{L}\phi \rangle. \quad (139)$$

(3.b) Show that $\mathcal{L}$ is positive, i.e.

$$\langle \psi, \mathcal{L}\psi \rangle \geq 0. \quad (140)$$

Also state under which condition equality holds.

(3.c) Explain why the statement that a differential operator is Hermitian always also depends on its domain of definition and on the chosen boundary conditions. What would change if boundary conditions other than the Dirichlet boundary condition were prescribed?

We now consider the corresponding vector version, as it occurs for electromagnetic eigenmodes in an ideally conducting cavity. For complex vector fields, let

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_V d^3\mathbf{r} \mathbf{f}^*(\mathbf{r}) \cdot \mathbf{g}(\mathbf{r}). \quad (141)$$

The boundary condition for an ideally conducting boundary is

$$\hat{\mathbf{n}} \times \mathbf{f}|_{\partial V} = 0. \quad (142)$$

Now consider the operator

$$\mathcal{M}\mathbf{f} = \nabla \times (\nabla \times \mathbf{f}). \quad (143)$$

(3.d) Show, using the identity

$$\nabla \cdot (\mathbf{f} \times \mathbf{g}) = (\nabla \times \mathbf{f}) \cdot \mathbf{g} - \mathbf{f} \cdot (\nabla \times \mathbf{g}) \quad (144)$$

as well as the divergence theorem, that $\mathcal{M}$ is Hermitian under the boundary condition

$$\hat{\mathbf{n}} \times \mathbf{f}|_{\partial V} = 0 \quad (145)$$

(3.e) Finally, show that $\mathcal{M}$ is positive:

$$\langle \mathbf{f}, \mathcal{M}\mathbf{f} \rangle = \int_V d^3\mathbf{r} |\nabla \times \mathbf{f}|^2 \geq 0. \quad (146)$$

Solution:

(3.a) We show that

$$\mathcal{L} = -\nabla^2 \quad (147)$$

is Hermitian, i.e.

$$\langle \mathcal{L}\psi, \phi \rangle = \langle \psi, \mathcal{L}\phi \rangle \quad \forall \psi, \phi. \quad (148)$$

$$\begin{aligned} \langle \mathcal{L}\psi, \phi \rangle &= \int_V d^3\mathbf{r}' (-\nabla^2 \psi^*(\mathbf{r}')) \phi(\mathbf{r}') \\ &= \int_V d^3\mathbf{r}' (-\nabla \cdot (\nabla \psi^*)) \phi(\mathbf{r}') \\ &= \int_V d^3\mathbf{r}' [\nabla \phi \cdot \nabla \psi^* - \nabla \cdot ((\nabla \psi^*) \phi(\mathbf{r}'))] \\ &= \int_V d^3\mathbf{r}' \nabla \phi \cdot \nabla \psi^* + \underbrace{\int_{\partial V} d\mathbf{A}' \cdot (\nabla \psi^*) \phi(\mathbf{r}')}_{=0} \\ &= \int_V d^3\mathbf{r}' [\nabla \cdot (\psi^* \nabla \phi) - \psi^* \nabla \cdot (\nabla \phi)] \\ &= \underbrace{\int_{\partial V} d\mathbf{A}' \cdot \psi^* \nabla \phi}_{=0} - \int_V d^3\mathbf{r}' \psi^* \Delta \phi = \langle \psi, \mathcal{L}\phi \rangle. \end{aligned} \quad (149)$$

The boundary terms vanish here because of the boundary conditions $\psi|_{\partial V} = \phi|_{\partial V} = 0$.

(3.b) Set $\psi = \phi$ in part (a). The result of the first integration by parts is then

$$\langle \mathcal{L}\psi, \psi \rangle = \int_V d^3\mathbf{r}' \nabla \psi \cdot \nabla \psi^* = \int_V d^3\mathbf{r}' |\nabla \psi|^2 \geq 0. \quad (150)$$

(3.c) In part (a), we always used that

$$\psi|_{\partial V} = 0. \quad (151)$$

Otherwise, the boundary terms do not vanish in general, and

$$\tilde{\Delta} \equiv \langle \mathcal{L}\psi, \phi \rangle - \langle \psi, \mathcal{L}\phi \rangle \quad (152)$$

depends on ψ, ϕ and on the boundary conditions.

(3.d) We want to show that

$$\langle \mathcal{M}\mathbf{f}, \mathbf{g} \rangle = \langle \mathbf{f}, \mathcal{M}\mathbf{g} \rangle \quad (153)$$

for all $\mathbf{f}, \mathbf{g}$ with

$$\hat{\mathbf{n}} \times \mathbf{f}|_{\partial V} = 0, \quad \hat{\mathbf{n}} \times \mathbf{g}|_{\partial V} = 0. \quad (154)$$

$$\begin{aligned}
\langle \mathcal{M}\mathbf{f}, \mathbf{g} \rangle &= \int_V d^3\mathbf{r}' (\nabla \times (\nabla \times \mathbf{f}^*)) \cdot \mathbf{g} \\
&= \int_V d^3\mathbf{r}' [\nabla \cdot ((\nabla \times \mathbf{f}^*) \times \mathbf{g}) + (\nabla \times \mathbf{f}^*) \cdot (\nabla \times \mathbf{g})] \\
&= \underbrace{\int_{\partial V} d\mathbf{A}' \cdot ((\nabla \times \mathbf{f}^*) \times \mathbf{g})}_{=0 \text{ since } \hat{\mathbf{n}} \times \mathbf{g}=0} + \int_V d^3\mathbf{r}' (\nabla \times \mathbf{f}^*) \cdot (\nabla \times \mathbf{g}) \\
&= \int_V d^3\mathbf{r}' [\nabla \cdot (\mathbf{f}^* \times (\nabla \times \mathbf{g})) + \mathbf{f}^* \cdot (\nabla \times (\nabla \times \mathbf{g}))] \\
&= \underbrace{\int_{\partial V} d\mathbf{A}' \cdot (\mathbf{f}^* \times (\nabla \times \mathbf{g}))}_{=0} + \int_V d^3\mathbf{r}' \mathbf{f}^* \cdot (\nabla \times (\nabla \times \mathbf{g})) \\
&= \langle \mathbf{f}, \mathcal{M}\mathbf{g} \rangle.
\end{aligned} \tag{155}$$

The boundary terms vanish here because of the boundary conditions $\hat{\mathbf{n}} \times \mathbf{f}|_{\partial V} = \hat{\mathbf{n}} \times \mathbf{g}|_{\partial V} = 0$.

(3.e) We set $\mathbf{g} = \mathbf{f}$, and hence after the first integration by parts in part (d) it follows that

$$\langle \mathcal{M}\mathbf{f}, \mathbf{f} \rangle = \int_V d^3\mathbf{r}' (\nabla \times \mathbf{f}^*) \cdot (\nabla \times \mathbf{f}) = \int_V d^3\mathbf{r}' |\nabla \times \mathbf{f}|^2 \geq 0. \tag{156}$$